Tutorial 8

Ex. 1

~~Objective~~, Step 1:

D_1: the cat chased the mouse

D_2: the dog barked at the cat

D_3: the mouse ran away from the cat.

Step 2:

Remove stop words: (the, at, away, from etc.)

D_1: cat chased mouse

D_2: dog barked cat

D_3: mouse ran cat

Step 3: Vocabularly

["barked", "cat", "chased", "dog", "mouse", "ran"]

Step 4: TF

Vocab	D_1	D_2	D_3
barked	0/3 = 0	1/3 = 0.33	0/3 = 0
cat	1/3 = 0.33	1/3 = 0.33	1/3 = 0.33
chased	1/3 = 0.33	0/3 = 0	0/3 = 0
dog	0/3 = 0	1/3	0/3 = 0
mouse	1/3 = 0.33	0/3 = 0	1/3 = 0.33
ran	0/3 = 0	0/3 = 0	1/3 = 0.33

Step 5: Normalize TF

→

Vocab	D_1	D_2	D_3
barked	0	$1+\log(0.33)$	0
cat	$1+\log(0.33)$	$1+\log(0.33)$	$1+\log(0.33)$
chased	$1+\log(0.33)$	0	0
dog	0	$1+\log(0.33)$	0
mouse	$1+\log(0.33)$	0	$1+\log(0.33)$
ran	0	0	$1+\log(0.33)$

$\downarrow$

Vocab	D_1	D_2	D_3
barked	0	0.518	0
cat	0.518	0.518	0.518
chased	0.518	0	0
dog	0	0.518	0
mouse	0.518	0	0.518
ran	0	0	0.518

Step 4 – IDF Calc.

Vocab	IDF

barked $= \log\left(\frac{N}{1+df_t}\right) \rightarrow \log_{10}\left(\frac{3}{1+1}\right) \rightarrow \log\left(\frac{3}{2}\right) \rightarrow 0.176$

cat $= \log_{10}\left(\frac{N}{1+df_t}\right) \rightarrow \log_{10}\left(\frac{3}{1+3}\right) \rightarrow \log\left(\frac{3}{4}\right) \rightarrow 0.124$

chased $= \log_{10}\left(\frac{N}{1+df_t}\right) \rightarrow \log_{10}\left(\frac{3}{1+1}\right) \rightarrow \log\left(\frac{3}{2}\right) \rightarrow 0.176$

dog $= \log_{10}\left(\frac{N}{1+df_t}\right) \rightarrow \log_{10}\left(\frac{3}{1+1}\right) \rightarrow \log_{10}\left(\frac{3}{2}\right) \rightarrow 0.176$

mouse $= \log_{10}\left(\frac{N}{1+dft}\right) \rightarrow \log_{10}\left(\frac{3}{1+2}\right) \rightarrow \log_{10}\left(\frac{3}{3}\right) \rightarrow 0$

ran $= \log_{10}\left(\frac{N}{1+dft}\right) \rightarrow \log_{10}\left(\frac{3}{1+1}\right) \rightarrow \log_{10}\left(\frac{3}{3}\right) \rightarrow 0.176$

Step 5: Compute TF

Vocab	D_1	D_2	D_3
barked	$0 \times 0.176 = 0$	0.091	0
cat	-0.064	-0.064	-0.064
chased	0.091	0	0
dog	0	0.091	0
mouse	0	0	0
ran	0	0	0.091

3. "word2vec"

3.1 Ex. 2 Data generation for CBOW

D_1: "The Cat chased the Mouse"
D_2: "The Dog Barked at the Cat."
D_3: "The mouse ran away from the cat"

After stop-words removal,
D_1: ["cat", "chased", "mouse"]
D_2: ["dog", "barked", "cat"]
D_3: ["mouse", "ran", "cat"]

Vocabulary:
["barked", "cat", "chased", "dog", "mouse", "ran"]

Step: 2 : Look-up table

dog barked cat

Context(words)	Context(Index)	Target(word)	Target(index)
[barked]	[0]	dog	[3]
[dog, cat]	[3, 1]	barked	[0]
[barked]	[0]	cat	[1]

mouse ran cat

Context(words)	Context(Index)	Target(word)	Target(index)
[ran]	[5]	mouse	[4]
[mouse, cat]	[4, 1]	ran	[5]
[ran]	[5]	cat	[1]

cat chased mouse

Context(words)	Context(index)	Target word	Target(index)
[chased]	[2]	cat	1
[cat, mouse]	[1, 4]	chased	2
[chased]	[2]	mouse	4